

Data Structures and Algorithms

Exercise 7: Financial Forecasting

CODE:

```
public class FinancialForecast {  
    public static double futureValue(double presentValue,  
                                     double growthRate,  
                                     int years) {  
        if (years == 0) {  
            return presentValue;  
        }  
        return futureValue(presentValue, growthRate, years - 1)  
            * (1 + growthRate);  
    }  
    public static void main(String[] args) {  
        double presentValue = 10000; // Initial investment  
        double growthRate = 0.08;    // 8% annual growth  
        int years = 5;  
        double result = futureValue(  
            presentValue,  
            growthRate,  
            years);  
        System.out.println("Future Value after "  
            + years + " years = "  
            + result);  
    }  
}
```

OUTPUT:

Output

Future Value after 5 years = 14693.280768000004

=== Code Execution Successful ===